Ex.No: 03 Setting up a Local Area Network (LAN)

Aim:

To design, set up and configure a LAN connecting and computers through a network switch. Assign IP addresses to each device, verify connectivity using ping command and demonstrate resource/data sharing across the network.

Equipment Required:

S.No	Equipments	Quantity	Specifications
1.	Personal computer (PC)	4	Ethernet (NIC enables
2.	Network Switch	1	8/16/24 Port
3.	Ethernet Cables (Cat5e/Cat6)	1	Straight-thro RJ-45

Implementation in CISCO Packet Tracer:

Step 1: Place the devices

- Open Cisco Packet Tracer.
- From vertical category panel, select End Devices and drag 4 PCs onto the workspace.
- Select Switches and drag switch onto workspace.

Step 2: Cable the devices

- Click Connections → Select Copper Straight-Through cable.
- Click each PC, then switch and select an available FastEthernet to complete each connection.

Step 3: Assign IP addresses to PCs.

- Click PC → Desktop tab → IP configuration
- Enter static IP address and subnet Mask as per the IP plan for each PC.

Step 4: Configure the Switch Management IP

- Click on Switch → CLI tab.
- Enter CLI commands to assign a management IP to VLAN 1:

→ enable
→ configure terminal
→ interface vlan 1
→ ip address 192.168.1.254 255.255.255.0
→ no shutdown
→ exit.

Step 5: Verify connectivity using Ping.

- Click on PC0 → Desktop tab → Command Prompt
- Use ping command to test connectivity to another PC.
- Successful "Reply from ..." messages confirm connectivity between devices.

Step 6:

- Add a Web Server and connect to switch
- Assign an IP address to the Server via Desktop → IP Configuration tab
- Open the service tab → Select HTTP/HTTPS tab and enable the HTTP service.

Step 7: Observe Packet flow in Simulation Mode

- Switch to Simulation Mode.
- Send a ping or web request between two devices and use the play controls to step through the packets as it travels from host through the switch, observing MAC address-based forwarding.